

**COMS 3110: Homework 5**  
**Due: December 6<sup>th</sup>, 11:59pm**  
**Total Points: 60**

**Late submission policy.** An assignment that is submitted one day late will get a penalty of 10%. An assignment that is submitted two days late will get a penalty of 20%. Assignments submitted after two days will not be graded and will get no points. For example, if an assignment is due on Friday by midnight, a submission on Saturday will be penalized by 10%, and a submission on Sunday will be penalized by 20%. A submission on Monday will get no points.

**Submission format.** Homework solutions will have to be typed. You can use word, LaTeX, or any other type-setting tool to type your solution. Your submission file should be in pdf format. Do **NOT** submit a photocopy of handwritten homework except for diagrams that can be hand-drawn and scanned. We reserve the right **NOT** to grade homework that does not follow the formatting requirements. Name your submission file: `<Your-net-id>-3110-hw5.pdf`. For instance, if your netid is `asterix`, then your submission file will be named `asterix-3110-hw5.pdf`. Each student must hand in their own assignment. If you discussed the homework or solutions with others, a list of collaborators must be included with each submission. Each of the collaborators has to write the solutions in their own words (copies are not allowed).

### General Requirements

- When proofs are required, do your best to make them both clear and rigorous. Even when proofs are not required, you should justify your answers and explain your work.
- When asked to present a construction, you should show the correctness of the construction.

### Some Useful (in)equalities

- $\sum_{i=1}^n i = \frac{n(n+1)}{2}$
- $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$
- $2^{\log_2 n} = n$ ,  $a^{\log_b n} = n^{\log_b a}$ ,  $n^{n/2} \leq n! \leq n^n$ ,  $\log x^a = a \log x$
- $\log(a \times b) = \log a + \log b$ ,  $\log(a/b) = \log a - \log b$
- $a + ar + ar^2 + \dots + ar^{n-1} = \frac{a(r^n - 1)}{r - 1}$
- $1 + \frac{1}{2} + \frac{1}{2^2} + \dots + \frac{1}{2^n} = 2(1 - \frac{1}{2^{n+1}})$
- $1 + 2 + 4 + \dots + 2^n = 2^{n+1} - 1$

1. **(10 pts)** You are helping a group of ethnographers analyze some oral history data. The ethnographers interviewed members of a village and learned about the birth and death a set of  $n$  people:  $P_1, P_2, \dots, P_n$ . The information is categorized as follows:

- For some  $i$  and  $j$ , person  $P_i$  died before person  $P_j$  was born
- For some  $i$  and  $j$ , the life spans of  $P_i$  and  $P_j$  overlapped at least partially.

As these are word of mouth information, it becomes important to validate the data. You are tasked with writing an algorithm which takes as input the learned information (about relative ordering of birth and death of  $n$  people) and verify its validity.

2. **(10 pts)** Solve the following recurrence equations using the recurrence tree method. Show the table as done in lecture.

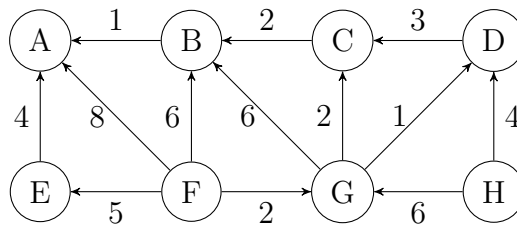
(a) **(5 pts)**

$$T(n) = \begin{cases} 1 & \text{if } n < 2 \\ 4 \cdot T(\frac{n}{2}) + n & \text{if } n \geq 2 \end{cases}$$

(b) **(5 pts)**

$$T(n) = \begin{cases} 1 & \text{if } n < 3 \\ T(\frac{n}{3}) + n & \text{if } n \geq 3 \end{cases}$$

3. **(10 pts)** Recall the Dijkstra's algorithm as presented in class. Consider the following weighted graph and write the valuation of  $d$ -value for each vertex at the end of each iteration of while-loop (that iterates until the priority queue is empty) in the algorithm. The source vertex is H.



(Add more rows to the following table, if needed, for your solution.)

|                    |          |          |          |          |          |          |          |   |
|--------------------|----------|----------|----------|----------|----------|----------|----------|---|
| Initial $d$ -value | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ | 0 |
| Iteration          | A        | B        | C        | D        | E        | F        | G        | H |
| 1                  |          |          |          |          |          |          |          |   |
| 2                  |          |          |          |          |          |          |          |   |
| 3                  |          |          |          |          |          |          |          |   |
| 4                  |          |          |          |          |          |          |          |   |
| 5                  |          |          |          |          |          |          |          |   |
| 6                  |          |          |          |          |          |          |          |   |
| 7                  |          |          |          |          |          |          |          |   |
| 8                  |          |          |          |          |          |          |          |   |

4. (10 pts) Prove the following claim: for any cycle in a weighted graph with distinct **non-negative** edge weights, the edge with the highest weight in the cycle does not belong to a MST.
5. (20 pts) Given two strings, write an algorithm for finding the longest common subsequence. A subsequence of a string is a sequence of characters which conforms to the relative ordering of the characters in the string. The characters do not necessarily appear contiguously in the string.

Example: GHTCCHT and CHTGCH. The longest common subsequence is HTCH.

Design an algorithm using a Dynamic Programming strategy which consists of the following:

- (7 pts) Formalize a recursive definition for the solution.
- (10 pts) Write pseudocode for an iterative dynamic programming algorithm with a dictionary to implement the solution.
- (3 pts) Analyze the runtime of your algorithm.